Phase Equilibrium Concentration Derivation for Binary Alloy Systems

Notes on Thermodynamics and Materials Science

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1 Known Conditions and Symbol Definitions

Consider a binary alloy system containing two solute elements (denoted as elem 1 and elem 2). The physical parameters are defined as follows.

1.1 Initial State

- Initial concentrations: C_1^0 (elem 1), C_2^0 (elem 2)
- Initial temperature: T_0 (liquidus temperature of the base matrix)

1.2 Physical Parameters

- Liquidus slopes: m_1 (elem 1), m_2 (elem 2) [K/at.%]
- Equilibrium partition coefficients: k_1 (elem 1), k_2 (elem 2) [dimensionless]
- Temperature correction term: ΔT_R (known constant accounting for matrix liquidus shift)

1.3 Current State (at a given temperature T)

- Actual liquid concentrations: C_1, C_2

1.4 Quantities to Solve

- Equilibrium interface concentrations: C_1^{eq}, C_2^{eq}
- Solid fraction increment: Δf_S

2 Governing Equations

Two core equations are established based on the linear thermodynamic approximation and the solute redistribution principle.

2.1 Equation 1: Temperature–Concentration Linearization

The local liquidus temperature is assumed to shift linearly with solute concentration, with contributions from each element superimposed:

$$T = T_0 + m_1\big(C_1^{eq} - C_1^0\big) + m_2\big(C_2^{eq} - C_2^0\big) + \Delta T_R. \tag{1}$$

2.2 Equation 2: Solid Fraction from Partition Coefficients

Based on the lever rule and the definition of the partition coefficient, the solid fraction increment Δf_S must be consistent for both elements:

$$\Delta f_S = \frac{C_1^{eq} - C_1}{C_1^{eq}(k_1 - 1)} = \frac{C_2^{eq} - C_2}{C_2^{eq}(k_2 - 1)}. \tag{2}$$

3 Derivation

3.1 Step 1: Relating C_1^{eq} and C_2^{eq} via Equation (2)

Cross-multiplying Eq. (2):

$$C_2^{eq}(k_2-1)(C_1^{eq}-C_1) = C_1^{eq}(k_1-1)(C_2^{eq}-C_2).$$

Let $A_i = k_i - 1$ for brevity. Expanding and rearranging yields an expression for C_2^{eq} as a function of C_1^{eq}:

$$C_2^{eq} = \frac{A_1\, C_2\, C_1^{eq}}{A_1\, C_1^{eq} - A_2(C_1^{eq} - C_1)}. \tag{3}$$

Substituting $A_1 = k_1 - 1$ and $A_2 = k_2 - 1$:

$$C_2^{eq} = \frac{(k_1-1)\, C_2\, C_1^{eq}}{(k_2-1)\, C_1^{eq} + (k_1-k_2)\, C_1}. \tag{4}$$

3.2 Step 2: Substituting into Equation (1)

Define the effective undercooling

$$\Delta T \;=\; T - T_0 - m_1 C_1^0 - m_2 C_2^0 - \Delta T_R.$$

Substituting Eq. (4) into Eq. (1) and letting the denominator $D = (k_2-1)\, C_1^{eq} + (k_1-k_2)\, C_1$, after multiplying through by D and collecting terms in C_1^{eq}, one obtains the quadratic equation:

$$a\left(C_1^{eq}\right)^2 + b\, C_1^{eq} + c = 0. \tag{5}$$

4 Closed-Form Results

4.1 Quadratic Coefficients

$$a = m_1(k_2-1) + m_2(k_1-1)\,\frac{C_2}{C_1}, \tag{6}$$

$$b = m_1(k_1-k_2)\, C_1 + m_2(k_1-1)\, C_2 - \Delta T\,(k_2-1), \tag{7}$$

$$c = -\Delta T\,(k_1-k_2)\, C_1, \tag{8}$$

where

$$\Delta T = T - T_0 - m_1 C_1^0 - m_2 C_2^0 - \Delta T_R.$$

4.2 Solution for C_1^{eq}

$$\boxed{C_1^{eq} = \frac{-b + \sqrt{b^2 - 4ac}}{2a}} \tag{9}$$

Remark. The positive root is selected because concentrations must be positive (typically $C_1^{eq} > C_1$ during solidification). If $a \to 0$ (i.e. $k_1 \approx k_2$), the equation degenerates to a linear problem solved directly from Eq. (7).

4.3 Solution for C_2^{eq}

$$\boxed{C_2^{eq} = \frac{(k_1 - 1)\, C_2\, C_1^{eq}}{(k_2 - 1)\, C_1^{eq} + (k_1 - k_2)\, C_1}} \tag{10}$$

4.4 Solid Fraction Increment

$$\boxed{\Delta f_S = \frac{C_1^{eq} - C_1}{C_1^{eq}(k_1 - 1)}} \tag{11}$$

5 Parameter Set: CoCrNi-Ti-C System

This section extracts parameters for the CoCrNi-Ti-C alloy system from the report: *"3D Cellular Automaton Simulation of Dendrite Growth in CoCrNi-Ti-C Alloys: A Comprehensive Parameterization Framework"* (March 2026).
These parameters are used to populate m_1, m_2, k_1, k_2, and related quantities in the binary derivation above.

5.1 System Definition

The CoCrNi-Ti-C system is treated as a pseudo-binary system:

- elem 1 = Titanium (Ti)
- elem 2 = Carbon (C)
- Matrix = equiatomic CoCrNi (effective solvent)

5.2 Thermodynamic Parameters

Table 1: Thermodynamic parameters (m_i, k_i) for the CoCrNi-Ti-C system.

Symbol	Description	Ti (elem 1)	C (elem 2)	Source
k_i	Equilibrium partition coefficient	0.06–0.12	0.03–0.08	CALPHAD (TCHEA6);
m_i	Liquidus slope (K/at.%)	−20 to −35	−40 to −80	CALPHAD; [9,13,15]
T_0	CoCrNi matrix liquidus temperature	1610 K (1337 °C)		Exp. DSC + CALPHAD
T_{sol}	Solidus temperature	1443 K (1170 °C)		Exp. DSC + CALPHAD

Notes:

- $k < 1$: both Ti and C strongly partition into the liquid phase.
- $m < 0$: solute enrichment depresses the liquidus temperature.
- Precise values should be extracted from Thermo-Calc + TCHEA6 at the exact composition.

5.3 Temperature Correction Term ΔT_R

For a given alloy $(\text{CoCrNi})_{1-2x}\text{Ti}_x\text{C}_x$, the liquidus temperature shifts relative to the pure CoCrNi matrix:

$$\Delta T_R = (m_1 + m_2)\, x.$$

For the representative composition $x = 0.004$ (0.4 at.% Ti and C each):

$$\Delta T_R \approx (-25 + (-60)) \times 0.4 = -34 \text{ K}.$$

Table 2: Initial alloy concentrations.

Symbol	Description	Typical value	Unit
C_1^0	Ti initial concentration	0.4 (range 0.2–1.0)	at.%
C_2^0	C initial concentration	0.4 (equimolar with Ti)	at.%

Table 3: Diffusion coefficients in liquid and solid phases.

Symbol	Description	Ti	C	Unit	Source
D_L	Liquid diffusivity	3×10^{-9}	8×10^{-9}	m^2/s	[13,16]
D_S	Solid diffusivity	≈ 0 (Scheil)	10^{-14}–10^{-13}	m^2/s	DICTRA (MOBHEA2) [9]

5.4 Initial Concentrations

5.5 Diffusion Kinetics Parameters

Note. Due to the sluggish diffusion effect in CoCrNi, solid back-diffusion D_S for Ti is safely neglected (Scheil–Gulliver approximation).

5.6 Thermophysical Properties

Table 4: Additional thermophysical parameters.

Symbol	Parameter	Value	Unit	Source
L	Latent heat of fusion	2.35×10^5	J/kg	Exp. [10]
ρ	Density	8100–8300	kg/m^3	[21]
c_p	Specific heat capacity	530–560	J/(kg·K)	[10]
σ_{SL}	Solid–liquid interfacial energy	0.18–0.25	J/m^2	Phase-field inversion [17]
Γ	Gibbs–Thomson coefficient	$(1–3) \times 10^{-7}$	K·m	[18]
ε_4	Anisotropy strength	0.03–0.05	–	[13]

5.7 Nucleation Parameters (for CET Modeling)

5.8 Recommended Parameter Values for the Binary Derivation

Substituting the above into the equations of Sections 2–4:

$$
\begin{aligned}
m_1 &= -25 \text{ K/at.\%} \quad \text{(Ti liquidus slope, recommended)} \\
m_2 &= -60 \text{ K/at.\%} \quad \text{(C liquidus slope, recommended)} \\
k_1 &= 0.08 \quad \text{(Ti partition coefficient, recommended)} \\
k_2 &= 0.05 \quad \text{(C partition coefficient, recommended)} \\
T_0 &= 1610 \text{ K} \quad \text{(CoCrNi matrix liquidus)} \\
\Delta T_R &= \text{from CALPHAD, or approximated as above}
\end{aligned}
$$

Table 5: Nucleation parameters for the columnar-to-equiaxed transition (CET) model.

Parameter	Description	Recommended value	Unit
$n_{\max,v}$	Volumetric nucleation density	10^{11}–10^{13}	m^{-3}
$\Delta T_{N,v}$	Mean volumetric nucleation undercooling	1.5–4	K
$\sigma_{\Delta T,v}$	Std. dev. of nucleation distribution	0.5–1	K
$n_{\max,s}$	Surface nucleation density	10^{12}–10^{14}	m^{-2}
$\Delta T_{N,s}$	Surface nucleation undercooling	≈ 0.5	K

The solved quantities C_1^{eq} and C_2^{eq} represent the equilibrium liquid concentrations of Ti and C at the solid–liquid interface; Δf_S is the solid fraction increment at each time step, reflecting the advancement rate of the solidification front.

Parameter source: "3D CA Simulation of Dendrite Growth in CoCrNi-Ti-C Alloys," March 2026.